

**alphabet**

**det**

**inversion-fixed points**

**who did it**

$(X, \bar{X})$

+1

none

Ciucu-Krattenthaler 2008

$(X, \bar{X}, 1)$

+1

+1: constant row

Ayyer-Behrend 2018

$(X, \bar{X}, 1, -1)$

-1

-1: parity row

*this paper*

the block identity needs every fixed point to give a CONSTANT row;  
-1 gives  $((-1)^{\mu_j})_j$ , and the triangularisation stops